HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
School of Information and Communications Technology

PROJECT TOPIC PROPOSAL

COMPUTER VISION – IT4343E

Fundamental Concepts and Methods in Biomedical Image Registration

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Chapter 1

Introduction

Biomedical image registration is an important problem in medical image analysis. It deals with the task of matching two or more medical images so that the same anatomical locations correspond to each other in space. These images may come from the same patient or from different patients. They may also be taken at different times, from different viewing positions, or with different imaging techniques. The main goal of registration is to align these images into one shared coordinate system so that the same organ, tissue, or anatomical point appears in the correct matching location across images [1, 2]. In practice, this alignment step is very important because it allows information from multiple scans to be combined. It is often needed before carrying out other medical image analysis tasks, such as segmentation, quantitative measurement, atlas building, motion tracking, image-guided treatment, and monitoring how a disease changes over time [1, 3]. A general overview of this process is illustrated in Figure 1.1.

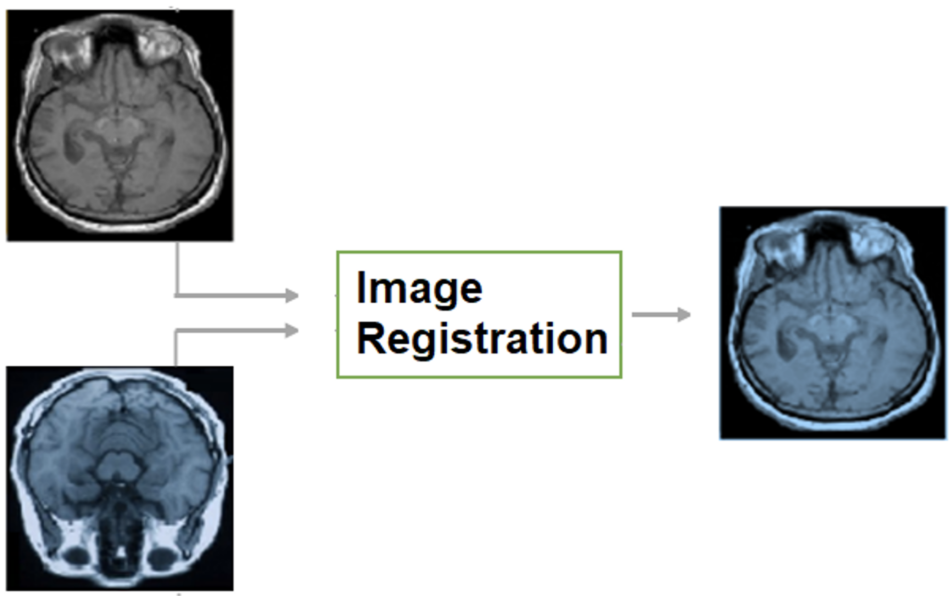


Figure 1.1: Conceptual workflow of biomedical image registration[1]

[1]Source: Ahmad Hammoudeh, Stéphane Dupont. (2023). Deep Learning in Medical Image Registration: Introduction and Survey. Qeios.

Image registration is necessary because medical images are rarely perfectly aligned by themselves. Even when scanning the same patient more than once, the images may look different for many reasons. The patient may lie in a slightly different position, the scanner settings may change, the scanning angle or geometry may be different, breathing may cause motion, soft tissues may deform, or the disease itself may progress between scans. When images come from different patients, the differences are usually even larger because human anatomy naturally varies from person to person. In addition, when different imaging modalities are used, each modality shows only part of the full biological picture. For example, computed tomography (CT) shows dense structures such as bones very clearly, magnetic resonance imaging (MRI) gives better contrast for soft tissues, and positron emission tomography (PET) shows metabolic or functional activity, as illustrated in Figure 1.2. Registration helps bring these different sources together into one consistent view, making the final result more useful than relying on a single image alone [3, 4].

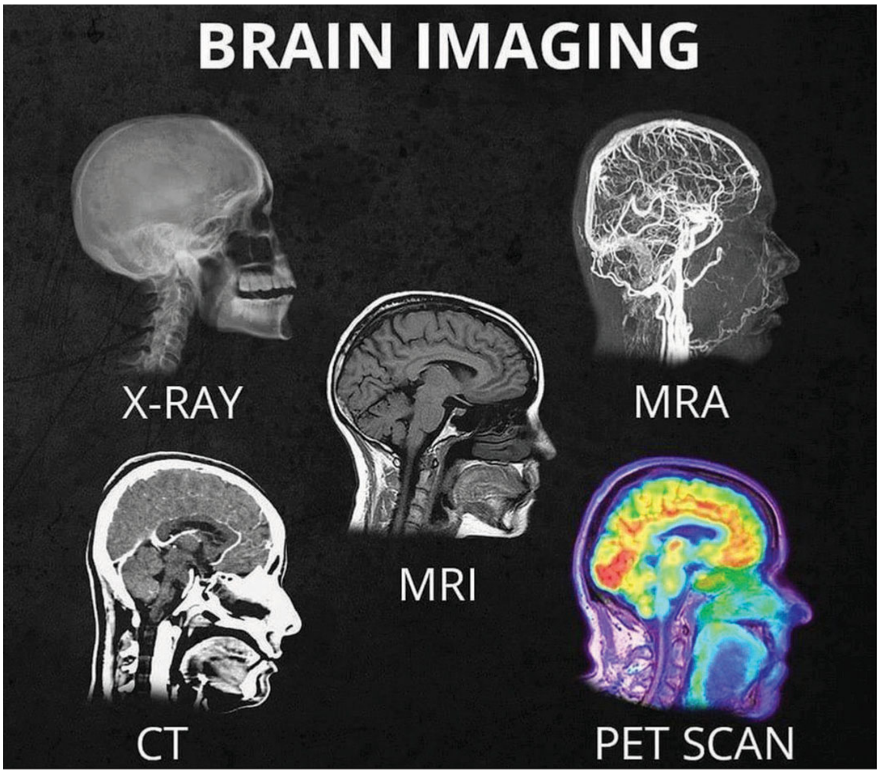


Figure 1.2: Comparison of brain imaging modalities: X-ray, CT, MRI, MRA, and PET[2]

The practical value of image registration can be seen in many biomedical and clinical applications. In multimodal image fusion, accurate alignment of CT, MRI, PET, or ultrasound images helps doctors combine structural information with functional information, which supports diagnosis and treatment planning. In longitudinal analysis, registration is needed to compare images taken at different times, for example to follow tumor growth, observe brain degeneration, or measure how well a treatment is working. In image-guided surgery and radiotherapy, scans taken before treatment or surgery must often be aligned with images taken during or after the procedure. This alignment helps with navigation, locating the treatment target, and adjusting the intervention when conditions change. In population-based studies, registration is also a key tool for building

[2]Source: San Diego Brain Injury Foundation (SDBIF). Detailed comparison available at sdbif.org

anatomical atlases, comparing different subjects, transferring labels from one image to another, and studying shape differences across groups of patients [1, 3, 5].

From the methodological point of view, biomedical image registration has developed significantly over time. Earlier approaches mainly used classical optimization methods. In these methods, researchers designed similarity measures by hand, defined explicit transformation models, and then used iterative numerical procedures to find the best alignment. Important examples include mutual-information-based multimodal registration, free-form deformation methods, and diffeomorphic frameworks such as DARTEL and SyN [4–7]. In recent years, deep learning has changed the field in a major way. Instead of solving a new optimization problem for every image pair during testing, learning-based methods train neural networks to predict deformation fields directly. This change has greatly improved speed and made large-scale applications more practical. Examples include convolutional neural network (CNN) methods such as VoxelMorph and Transformer-based methods such as TransMorph [3, 8, 9].

Even with these advances, image registration is still difficult and is considered an inherently ill-posed problem. This means that there may be more than one possible transformation that seems to align the images well, especially when image quality is poor, when it is hard to determine correct correspondences, or when the two images overlap only partially. The problem becomes even more challenging in multimodal registration, because the same anatomical structure may appear very different in intensity or texture across imaging modalities. It is also difficult in deformable registration, where the transformation must handle large non-rigid motion while still remaining realistic from an anatomical point of view. Because of these challenges, topics such as regularization, preservation of topology, and robust similarity modeling are central throughout the literature [1, 3–5].

This project focuses on the basic ideas and the main methodological families in biomedical image registration. In particular, it emphasizes four important strategies: classical optimization-based methods, metaheuristic methods, CNN-based registration, and Transformer-based registration. Before reviewing these methods in detail, it is necessary to define the registration problem carefully and explain the mathematical roles of the transformation model, the similarity term, regularization, and geometric constraints. For this reason, the next chapter presents a detailed overview of the problem and provides the conceptual foundation for the rest of the report.

Chapter 2

Problem Overview

2.1 Image Registration Problem Definition

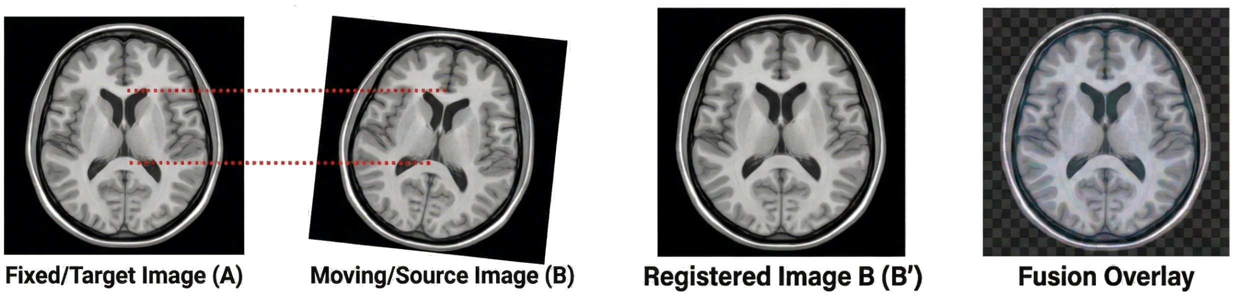
Let $\Omega \subset \mathbb{R}^d$ denote the image domain, where $d = 2$ for planar images and $d = 3$ for volumetric scans. Let

$$F : \Omega \to \mathbb{R} \quad \text{and} \quad M : \Omega \to \mathbb{R} \tag{2.1}$$

denote the *fixed image* and the *moving image*, respectively. The goal of image registration is to estimate a spatial transformation

$$\phi : \Omega \to \Omega \tag{2.2}$$

such that the transformed moving image $M \circ \phi$ is aligned with the fixed image F [1, 5]. This basic input-output structure of the registration problem is illustrated in Figure 2.1.



(a) (b)

Figure 2.1: Input/Output of Biomedical image registration problem:
(a) Input fixed image (A) and moving image (B) with spatial correspondences;
(b) Resulting registered image (B') and fusion overlay[1]

In other words, registration tries to find which location in one image matches which location in another image. If $x \in \Omega$ is a point in the coordinate system of the fixed image, then $\phi(x)$ gives the matching point in the moving image. After this mapping is defined, the moving image can be warped to the fixed image space as

$$(M \circ \phi)(x) = M(\phi(x)). \tag{2.3}$$

[1]Source: AI-generated image.

This means that, for each point x in the fixed image, we look at the value of the moving image at the corresponding location $\phi(x)$. In practice, however, $\phi(x)$ usually does not fall exactly on the discrete pixel or voxel grid of the moving image. Because of this, interpolation is needed to estimate the image value at that location. Common interpolation methods include nearest-neighbor interpolation, linear interpolation, and spline interpolation. Although interpolation may seem like only a technical detail, it can strongly affect both the stability of the optimization process and the final registration accuracy [1, 4].

The formulation above is written in a general way on purpose. It can describe rigid registration, affine registration, and deformable registration. It can also cover mono-modal and multi-modal registration, as well as both classical methods and modern learning-based methods. The main differences between these approaches come from three questions: how the transformation ϕ is represented, how similarity between images is measured, and how the optimization or learning procedure is carried out [1, 2].

2.2 Optimization Formulation

A common way to describe image registration is to treat it as an optimization problem, where the goal is to find the transformation that gives the best alignment between two images. This is often written as

$$\phi^* = \arg\min_{\phi \in \mathcal{T}} \Big[\mathcal{D}\big(F, M \circ \phi\big) + \lambda\, \mathcal{R}(\phi) \Big], \tag{2.4}$$

where $\mathcal{T}$ is the set of all transformations that are allowed, $\mathcal{D}$ is the similarity term that measures how well the transformed moving image matches the fixed image, $\mathcal{R}$ is the regularization term that controls how reasonable the transformation is, and $\lambda > 0$ is a weighting parameter that balances these two objectives [5, 6].

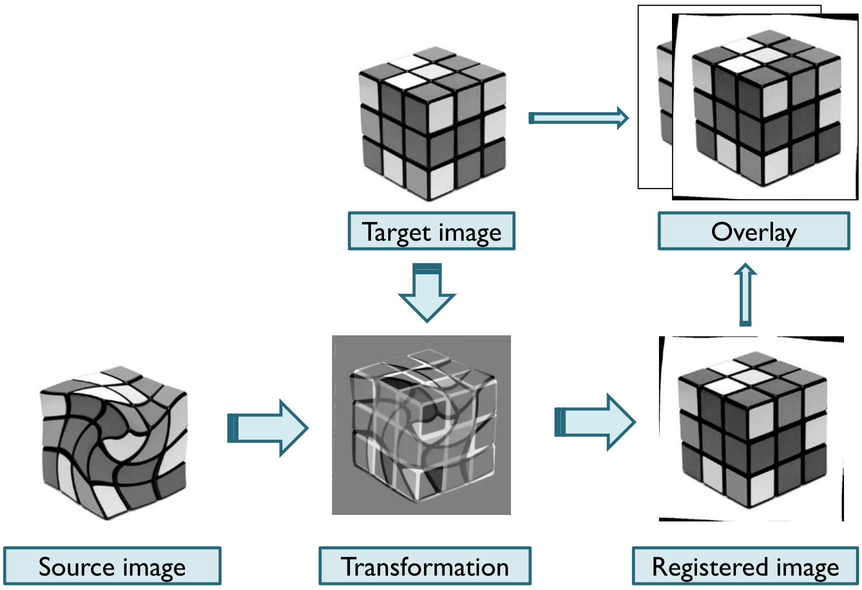


Figure 2.2: Diagram of image registration components[2]

[2]Source: BioOptics Facility, IMP. Available at: cores.imp.ac.at

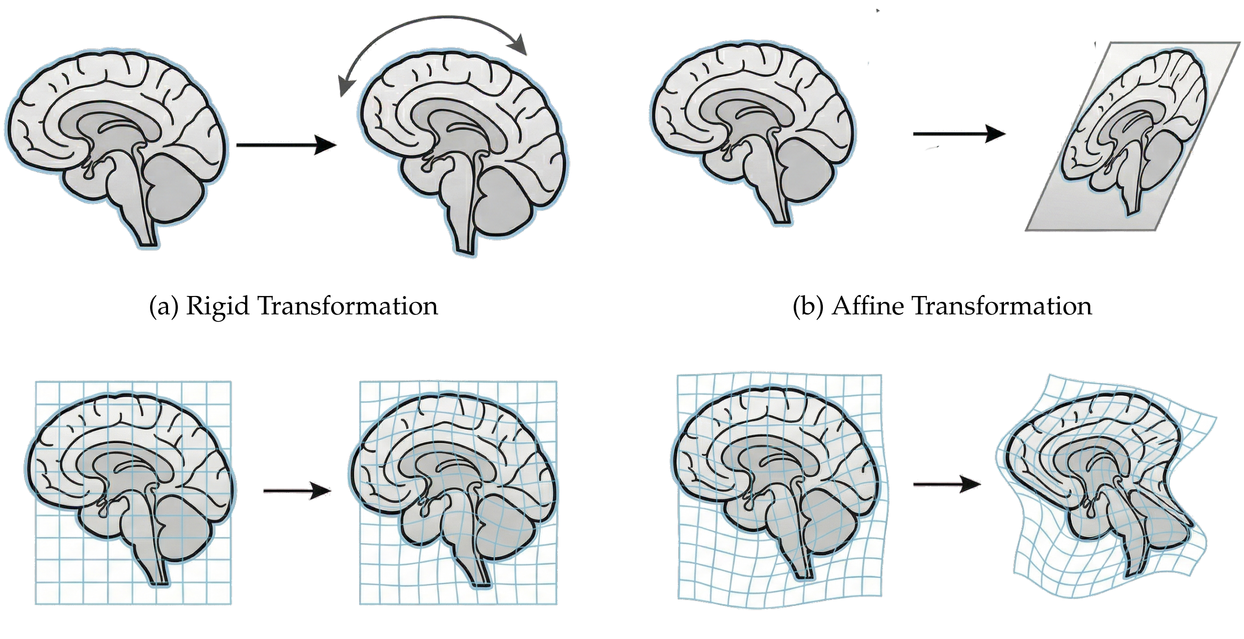
This formulation shows one of the main difficulties in image registration. On the one hand, the transformed image should match the target image as closely as possible. On the other hand, the transformation should still remain smooth, anatomically realistic, and, in many applications, invertible. In other words, the transformation should not only fit the images well, but should also make sense from a physical or anatomical point of view. If the similarity term is given too much importance, the method may fit image noise or create unrealistic local deformations just to improve the apparent match. If the regularization term is too strong, the transformation may become too restricted and may fail to capture real anatomical changes or natural variability [3, 6, 7].

For parametric models, the transformation is often written as ϕ_θ, where θ is a finite-dimensional vector of parameters.Then the registration problem can be rewritten as

$$\theta^* = \arg\min_{\theta} \Big[\mathcal{D}\big(F, M \circ \phi_\theta\big) + \lambda\,\mathcal{R}(\phi_\theta)\Big]. \tag{2.5}$$

This form is common in rigid and affine registration, where the unknown parameters are usually a small set of values such as rotation, translation, scaling, and shearing. In deformable registration, however, the number of parameters is often much larger, because the transformation must describe local changes that vary across space rather than only one global motion for the whole image [1, 6].

2.3 Transformation Models



(a) Rigid Transformation

(b) Affine Transformation

(c) Deformable Transformation

(d) Diffeomorphic Transformation

Figure 2.3: Illustration of four transformation models in biomedical image registration: rigid, affine, deformable, and diffeomorphic transformations.[3]

2.3.1 Rigid Transformation

Rigid registration assumes that shape is preserved and only global position and orientation are different. In 3D, a rigid transformation has six degrees of freedom: three translations and three rotations. In other words, the object can move left-right, up-down, and forward-backward, and it can also rotate around three axes. As illustrated in Figure 2.3(a), this model preserves the overall shape of the anatomy while allowing only global motion. This model is appropriate when the anatomy behaves approximately like a rigid body, such as the skull, or when rigid alignment is used as an initial step before applying a more flexible method [1].

2.3.2 Affine Transformation

Affine registration extends the rigid model by allowing anisotropic scaling and shearing in addition to rotations and translations. As shown in Figure 2.3(b), the transformation can change the global geometry of the image while remaining a linear mapping. Here, anisotropic scaling means that the image can be stretched or compressed by different amounts along different directions, while shearing means that the shape can be slanted. A 3D affine transformation is commonly written as

$$\phi(x) = Ax + b, \tag{2.6}$$

where $A \in \mathbb{R}^{3\times 3}$ is a linear transformation matrix and $b \in \mathbb{R}^3$ is a translation vector. Affine models account for global geometric differences between scans, such as orientation mismatch or overall scale change, but they are still not enough for complex soft-tissue deformation, where different parts of the anatomy may move in different ways [1, 6].

2.3.3 Deformable Transformation

In biomedical imaging, deformable or non-rigid registration is often the most relevant setting because living tissue is elastic and anatomical variation is mainly local. As illustrated in Figure 2.3(c), deformable registration allows different regions to undergo different local displacements. A common form is

$$\phi(x) = x + u(x), \tag{2.7}$$

where $u : \Omega \to \mathbb{R}^d$ is a dense displacement field. This means that each spatial location has its own displacement vector, so different parts of the image can move differently. Each spatial location is allowed to move independently, subject to smoothness constraints [3, 5].

Deformable transformations can be represented in different ways. As shown in Figure 2.4, image registration in practice is often viewed as a process that takes a source image and a target image, estimates a transformation, and produces a registered image together with an overlay for visual comparison. Parametric non-rigid models, such as B-spline free-form deformations, describe the displacement field by a sparse set of control points and interpolate the motion between them. In other words, the deformation is directly specified only at selected points, and the motion at the remaining locations is obtained smoothly from those points. This greatly reduces the number of unknowns while

[3] Source: AI-generated illustrations created for this report.

preserving local flexibility [6]. Non-parametric models, in contrast, optimize directly over dense displacement or velocity fields. These formulations are more flexible but also more computationally expensive and depend more strongly on regularization [5, 7].

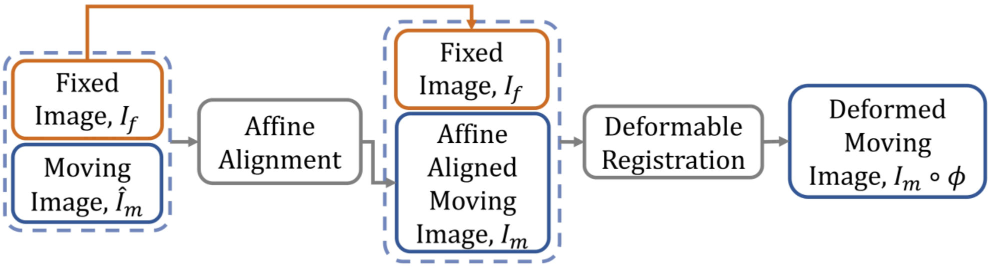


Figure 2.4: The conventional paradigm of image registration[4]

2.3.4 Diffeomorphic Transformation

In many medical applications, especially brain morphometry and atlas construction, the transformation is required to be diffeomorphic, that is, smooth, invertible, and topology preserving. As illustrated in Figure 2.3(d), this type of transformation allows complex deformation while maintaining a coherent anatomical structure. In simple terms, a diffeomorphic map can deform anatomy continuously without tearing, folding, or creating self-intersections. This means that neighboring anatomical points remain in a consistent order after deformation, which helps preserve the natural structure of the anatomy. This property is particularly important when registration is used not only for alignment, but also for analyzing anatomical shape differences themselves [5, 7].

2.4 Similarity Term

The similarity term $\mathcal{D}(F, M \circ \phi)$ measures how well the transformed moving image matches the fixed image. Its choice depends strongly on whether the two images come from the same modality.

2.4.1 Mono-modal Similarity

When the images are acquired with the same imaging modality and similar acquisition settings, corresponding structures are expected to have similar intensity patterns. In this setting, simple voxelwise measures are often effective. In other words, if two voxels represent the same anatomical structure, their intensity values are expected to be close.

A standard example is the sum of squared differences (SSD):

$$\mathcal{D}_{\mathrm{SSD}}(F, M \circ \phi) = \sum_{x \in \Omega} \big(F(x) - M(\phi(x))\big)^2. \tag{2.8}$$

[4]Source: J. Chen et al., "TransMorph: Transformer for unsupervised medical image registration," Medical Image Analysis, vol. 82, 2022.

SSD is simple and computationally efficient, but it assumes a direct intensity correspondence, meaning that matching structures should have nearly the same intensity values in both images. Because of this, SSD is sensitive to intensity non-uniformity, bias fields, and contrast changes [1].

Another widely used measure is normalized cross-correlation (NCC), which is more robust to linear intensity scaling. This means it can still work well when one image is overall brighter or darker than the other, as long as the relative intensity pattern remains similar. Conceptually, NCC rewards consistent relative intensity variation rather than exact intensity equality. For this reason, cross-correlation and its local variants are especially popular in brain MRI registration, including the SyN framework [5].

2.4.2 Multi-modal Similarity

In multi-modal registration, the same anatomy may have very different intensity values across modalities. For example, bone appears bright in CT but not necessarily in MRI, while PET primarily reflects metabolic uptake rather than tissue morphology. In such settings, direct intensity comparison becomes unreliable, and the relationship between image intensities must be modeled in a more general way [4].

The most influential measure in this setting is mutual information (MI), defined as

$$MI(F, M \circ \phi) = H(F) + H(M \circ \phi) - H(F, M \circ \phi), \tag{2.9}$$

where $H(\cdot)$ denotes entropy and $H(\cdot, \cdot)$ denotes joint entropy. Informally, entropy measures the amount of uncertainty or information in an image, while joint entropy measures the uncertainty when the two images are considered together. MI measures how much information one image contains about the other, and it became a standard solution for multimodal registration because it does not require a simple linear relationship between intensity values [4]. In practice, MI is often maximized, or equivalently its negative is minimized.

2.4.3 Learned Similarity

In modern deep learning-based registration, the similarity term may also be learned implicitly or explicitly through neural networks. Nevertheless, many recent models still rely on classical measures such as SSD, NCC, or local cross-correlation during training, especially in unsupervised settings. This shows that deep learning has not removed the basic registration objective. Instead, it has mainly changed how the transformation is estimated or predicted [3, 8, 9].

2.5 Regularization

The registration problem is inherently ill-posed: many different transformations may produce similar image similarity scores, especially in homogeneous regions or when noise is present. In other words, more than one transformation may seem to align the images equally well, even if some of them are not physically or anatomically reasonable. Regularization therefore plays a central role in limiting the set of possible solutions [1, 3].

2.5.1 Smoothness Regularization

A common regularizer penalizes rapid spatial variation in the displacement field:

$$\mathcal{R}_{\text{smooth}}(u) = \int_{\Omega} \|\nabla u(x)\|_2^2 \, dx. \tag{2.10}$$

where $u(x)$ describes how much each point moves, as described in section 2.3.3. The gradient $\nabla u(x)$ measures how quickly this displacement changes from one location to a nearby location. If neighboring points move in very different ways, then the gradient becomes large. By squaring its magnitude and integrating over the whole image domain Ω, the formula assigns a large penalty to transformations with abrupt spatial changes.

As a result, the optimization is encouraged to produce smoother motion fields in which nearby points move in similar directions and by similar amounts. This reflects the physical intuition that biological tissue usually deforms continuously rather than through sudden changes [6].

2.5.2 Bending Energy Regularization

For spline-based models, a higher-order regularizer known as bending energy is often used:

$$\mathcal{R}_{\text{bend}}(u) = \int_{\Omega} \sum_{i,j} \left(\frac{\partial^2 u(x)}{\partial x_i \partial x_j} \right)^2 dx. \tag{2.11}$$

While smoothness regularization looks at first-order change, bending energy looks at second-order change, that is, how quickly the change itself is changing. This is similar to the difference between a gently curved line and a sharply bent line. The second derivatives measure curvature in the displacement field, and the squared terms make large curvature contribute much more strongly to the penalty. Summing over all coordinate directions and integrating over the whole domain gives a global measure of how much the transformation bends. Therefore, this regularizer discourages deformation fields that bend too sharply, and is especially suitable when the transformation is expected to vary smoothly across space [6].

2.6 Geometric Constraints

Besides matching image intensities, a good registration method should also produce a transformation that is geometrically reasonable. In medical imaging, this means that the deformation should reflect realistic anatomical motion rather than an unrealistic mathematical distortion.

2.6.1 Topology Preservation

A desirable transformation should preserve the overall spatial arrangement of anatomical structures. Tissues that are close to each other should remain close after deformation, and the mapping should not create folding, tearing, or unrealistic overlap between regions. In practice, many methods check this behavior through the Jacobian determinant of the transformation:

$$J_{\phi}(x) = \det(\nabla \phi(x)). \tag{2.12}$$

This quantity describes the local volume change caused by the transformation around point x. If $J_\phi(x) > 0$, the local mapping keeps the original orientation; if $J_\phi(x) \leq 0$, folding or inversion may occur, which is generally considered anatomically unrealistic. For this reason, modern registration methods often try to avoid negative Jacobian values, either by adding explicit penalties or by using transformation models with stronger geometric guarantees [5, 7, 9].

2.6.2 Invertibility

In many applications, the estimated transformation should be invertible, meaning that correspondences can be followed consistently in both directions. Intuitively, if a point in one image is mapped to another image, it should also be possible to map it back in a meaningful way. This property is important in applications such as longitudinal analysis, atlas construction, and morphometric studies, where the transformation itself may later be studied as a representation of anatomical change [5, 7].

2.6.3 Anatomical Plausibility

More generally, the deformation field should remain consistent with the physical behavior of biological tissue. Real anatomical motion is usually smooth and spatially consistent rather than irregular or discontinuous. Therefore, a transformation that achieves high image similarity but produces unrealistic deformation is not desirable in practice. This is one of the main reasons why medical image registration requires not only a similarity term, but also constraints that encourage realistic geometric behavior [1, 3].

2.7 Classical Optimization View of Registration

In classical registration, the transformation is estimated separately for each image pair by solving the optimization problem directly. In other words, whenever two new images are given, the algorithm must start from the beginning and search for the best transformation for that pair alone. The procedure usually involves four interacting components: a transformation model, which determines what kinds of motion are allowed; a similarity metric, which measures how well the two images match after transformation; an optimizer, which searches for the best transformation; and a multi-resolution strategy, which makes the search more stable and efficient [1, 2].

The optimizer may be gradient descent, Gauss–Newton, Levenberg–Marquardt, or another numerical method, depending on the smoothness and dimensionality of the objective function. These methods can be understood as different ways of adjusting the transformation step by step in order to reduce the registration error. In practice, the optimization landscape is often non-convex, which means there may be many locally good solutions but not all of them correspond to the true anatomical alignment. Because of this difficulty, multilevel image pyramids are commonly used. The transformation is first estimated on coarse-resolution images, where fine details are removed and only the large overall structures remain, and is then gradually refined at higher resolutions. This coarse-to-fine strategy makes the method more robust to local minima and allows the algorithm to capture large deformations first before focusing on smaller details [5–7].

The main advantage of classical optimization is that it does not require training data and can often generalize well across different imaging settings. This is because it solves the registration problem directly for the given images rather than depending on patterns learned in advance from a dataset. Its main limitation is computational cost: the whole optimization process must be repeated from scratch for every new pair of images. This becomes a serious bottleneck in large-scale studies or in clinical workflows where results are needed quickly [3, 8].

2.8 Learning-Based View of Registration

Modern deep learning-based registration reformulates the problem by learning a function

$$f_\omega(F, M) \approx \phi, \tag{2.13}$$

where ω denotes the trainable parameters of a neural network. Instead of solving a new optimization problem for each test pair, the network is trained on many image pairs and then learns to predict the transformation directly. After training, registration can often be performed in a single forward pass through the network, which is much faster than iterative optimization [8, 9].

A common unsupervised learning objective is

$$\omega^* = \arg\min_{\omega} \sum_{(F,M)\in\mathcal{S}} \Big[\mathcal{D}\big(F, M \circ f_\omega(F, M)\big) + \lambda\, \mathcal{R}\big(f_\omega(F, M)\big)\Big], \tag{2.14}$$

where $\mathcal{S}$ is the training set. This objective is very similar to the classical variational formulation. The key difference is where the optimization takes place. In classical registration, optimization is performed separately for each image pair in order to find its transformation directly. In learning-based registration, optimization is performed during training in order to learn the network parameters, and after that the network can predict transformations for new image pairs without repeating the full iterative search [3, 8].

This perspective shows an important continuity between classical and modern methods. CNN-based and Transformer-based registration methods do not completely replace the classical formulation. Instead, they learn how to approximate its solution more efficiently. Put simply, the network learns from many examples how to produce transformations that would otherwise require a costly optimization procedure for each new pair of images. VoxelMorph is a representative CNN-based example of this idea, while TransMorph extends it by using Transformer blocks to model long-range spatial relationships more effectively, meaning that it can capture interactions between image regions that are far apart from each other [8, 9].

2.9 Discussion

Biomedical image registration is difficult for several reasons. First, in many cases there is no exact ground-truth correspondence between two images, especially in deformable registration. In other words, we usually do not know the true matching location for every point in the image. This makes both evaluation and supervised learning more difficult [3]. Second, the problem is often multimodal, which means the images come from different imaging techniques. In this case, the same anatomical structure may appear very different in intensity, so direct intensity comparison is not always reliable [4]. Third, anatomical motion can be large, local, and different for each organ. For example, soft tissue can deform in complex ways, and thoracic imaging is even more challenging because breathing causes motion and some organs may slide relative to nearby structures. Fourth, medical images may contain noise, artifacts, partial field-of-view mismatch, or disease-related changes, all of which make correct correspondence harder to determine [1, 3].

For these reasons, registration should not be seen as only an image matching task. It is better understood as a constrained inference problem, where the goal is to recover anatomically meaningful spatial correspondence from image data that may be incomplete or imperfect. This is why transformation design, similarity modeling, regularization, and prior knowledge are all important parts of a successful registration method.

In summary, biomedical image registration aims to estimate a transformation that aligns a moving image to a fixed image while balancing image matching with geometric and anatomical realism. The general objective includes a similarity term, a regularization term, and a transformation model, together with constraints such as smoothness, invertibility, and topology preservation. Classical methods solve this problem separately for each image pair using numerical optimization, while modern deep learning methods learn a mapping from image pairs to transformations and can therefore perform inference much faster. This mathematical framework provides the common basis for analyzing the four method families discussed in this report: classical optimization-based methods, metaheuristic methods, CNN-based methods, and Transformer-based methods.

Chapter 3

Methods

This project focuses on four representative method families in biomedical image registration: **classical optimization-based methods**, **metaheuristic methods**, **CNN-based methods**, and **Transformer-based methods**. These four groups capture the major evolution of the field from explicit variational optimization to learned deformation prediction. Although their implementation details differ, all four can ultimately be interpreted through the same general registration objective introduced earlier:

$$\phi^* = \arg\min_{\phi \in \mathcal{T}} \Big[\mathcal{D}(F, M \circ \phi) + \lambda \mathcal{R}(\phi)\Big]. \tag{3.1}$$

The main difference is how the transformation is parameterized and how the solution is obtained [1, 3, 8–10].

3.1 Classical Optimization-Based Registration

3.1.1 General Idea

Classical registration methods solve the registration problem directly as a numerical optimization problem for each image pair. Given a fixed image F and a moving image M, the method iteratively updates the transformation parameters so as to improve image similarity while maintaining geometric plausibility. A classical registration pipeline usually contains four key components:

1. a transformation model,
2. a similarity measure,
3. an optimizer,
4. and a regularization strategy.

This framework dominated medical image registration for many years and remains the conceptual foundation of the field [1, 10, 11].

In classical methods, the similarity term is typically hand-crafted. For mono-modal registration, common choices include SSD and NCC; for multi-modal registration, mutual information is the most influential choice. The transformation can be rigid, affine, or deformable, depending on the anatomical setting. The optimization is then carried out with gradient-based or variational methods, often under a multi-resolution coarse-to-fine strategy to reduce sensitivity to local minima [1, 4, 10].

3.1.2 Representative Methods: Demons and SyN

Among classical deformable registration methods, **Demons** and **Symmetric Normalization (SyN)** are particularly representative.

The Demons algorithm, originally inspired by optical flow, updates the displacement field based on local intensity differences and image gradients. It is computationally efficient and conceptually intuitive, and later diffeomorphic variants improved its mathematical consistency and topology-preserving behavior [12]. Demons became influential because it demonstrated that dense non-rigid correspondence could be estimated efficiently using iterative local updates.

SyN, proposed by Avants et al., is one of the most widely used classical deformable registration methods in medical imaging. It uses a symmetric diffeomorphic formulation, meaning that the mapping from image A to image B and the mapping from B to A are treated consistently through a midpoint-based optimization framework. SyN is especially well known for strong performance in brain MRI registration and has become a standard baseline in many later studies, including comparisons against deep learning methods [5].

3.1.3 Strengths and Limitations

The main advantage of classical optimization-based registration is that it does not require training data. As a result, these methods can be applied to new image pairs without prior learning and often generalize well across datasets and scanners. They are also easier to interpret, since every part of the objective function is explicitly defined.

However, this family has two major weaknesses. First, the optimization must be repeated from scratch for every new image pair, which can be computationally expensive. Second, the performance depends strongly on the choice of similarity measure, regularization, and initialization. In difficult settings such as large thoracic deformation or multimodal registration, local minima and slow convergence remain persistent challenges [3, 10].

3.2 Metaheuristic Registration

3.2.1 General Idea

Metaheuristic registration methods also formulate image registration as an optimization problem, but unlike classical gradient-based methods, they solve it using global search strategies inspired by biological evolution, collective behavior, or thermodynamic processes. Typical examples include genetic algorithms (GA), particle swarm optimization (PSO), differential evolution (DE), and simulated annealing (SA) [10, 13].

Instead of relying on gradient information, these methods maintain a set of candidate solutions and iteratively improve them according to stochastic search rules. If the transformation is parameterized by a vector θ, the objective becomes

$$\theta^* = \arg\min_{\theta} \Big[\mathcal{D}(F, M \circ \phi_\theta) + \lambda \mathcal{R}(\phi_\theta) \Big], \tag{3.2}$$

but the search for θ^* is now performed through population-based or probabilistic exploration rather than local gradient updates.

3.2.2 Representative Methods

Genetic Algorithms (GA). In GA-based registration, each candidate transformation is encoded as a chromosome. A population of candidate solutions evolves over generations through selection, crossover, and mutation. Solutions with better fitness values, typically defined using image similarity, are more likely to survive and produce offspring. GA is attractive because it can explore a large search space and is less sensitive to initialization than local optimization. However, it can be computationally expensive and may require careful tuning of mutation and crossover rates [10, 13].

Particle Swarm Optimization (PSO). PSO is one of the most common metaheuristics in registration. Each particle represents a candidate transformation and moves through parameter space according to both its own historical best position and the best position found by the swarm. The standard velocity-position update is

$$v_i^{(t+1)} = w v_i^{(t)} + c_1 r_1 \big(p_i^{best} - x_i^{(t)}\big) + c_2 r_2 \big(g^{best} - x_i^{(t)}\big), \tag{3.3}$$

$$x_i^{(t+1)} = x_i^{(t)} + v_i^{(t+1)}, \tag{3.4}$$

where $x_i^{(t)}$ and $v_i^{(t)}$ are the current position and velocity of particle i, p_i^{best} is its personal best solution, g^{best} is the global best solution found so far, and w, c_1, c_2, r_1, r_2 are control parameters [14, 15]. In registration, PSO is often combined with mutual information or feature-based similarity and is particularly useful when the objective surface is highly non-convex.

Differential Evolution (DE) and Simulated Annealing (SA). DE perturbs candidate solutions by combining differences between other candidates in the population, making it effective for continuous parameter optimization. SA, by contrast, probabilistically accepts worse intermediate solutions in order to escape local minima, gradually reducing this probability over time through a temperature schedule. Both methods have been applied to image registration, especially in linear or low-dimensional settings [10, 13].

3.2.3 When Metaheuristics Are Useful

Metaheuristic methods are most appealing when the objective function is highly non-smooth, contains many local minima, or lacks reliable gradient information. This often occurs in multi-modal registration, in feature-based formulations with discrete correspondences, or in low-dimensional rigid and affine alignment problems. Their global search ability can make them more robust than local gradient descent in such situations [10, 14].

3.2.4 Strengths and Limitations

The main strength of metaheuristics is their global exploration capability. They do not require differentiability and are less sensitive to initialization. This makes them useful when classical gradient-based methods fail or when similarity measures are noisy and highly irregular.

Their main weakness is computational cost. Population-based search is usually much slower than local optimization, especially in high-dimensional deformable registration. For dense 3D deformation fields, direct metaheuristic search becomes impractical unless combined with a low-dimensional parameterization or hybrid refinement strategy. Therefore, metaheuristics are often more suitable as complementary or specialized tools than as universal solutions for large-scale deformable registration [10, 13].

3.3 CNN-Based Registration

3.3.1 General Idea

CNN-based registration replaces iterative pairwise optimization with a learned mapping from image pairs to deformation fields. Instead of solving a new optimization problem for each test pair, a convolutional neural network is trained to predict the registration transformation directly:

$$\phi = f_\omega(F, M), \tag{3.5}$$

where ω denotes the trainable parameters of the network [8]. This approach is often described as *amortized optimization*: the expensive optimization is shifted from test time to training time.

In unsupervised or self-supervised CNN registration, the network is trained with a loss that mirrors the classical variational formulation:

$$\mathcal{L}(\omega) = \mathcal{D}\big(F, M \circ f_\omega(F, M)\big) + \lambda \mathcal{R}\big(f_\omega(F, M)\big). \tag{3.6}$$

Thus, the conceptual structure of the problem remains unchanged; only the inference mechanism changes.

3.3.2 Representative Method: VoxelMorph

The most influential CNN-based registration framework is **VoxelMorph**. VoxelMorph takes the fixed and moving images as input, concatenates them channel-wise, and passes them through an encoder-decoder convolutional architecture similar to a U-Net. The network outputs either a dense displacement field or a stationary velocity field, which is then used to warp the moving image through a differentiable spatial transformer module [8].

VoxelMorph was significant for two reasons. First, it demonstrated that unsupervised deformable registration could be learned end-to-end using only image similarity and deformation smoothness, without requiring ground-truth deformation fields. Second, it showed that once trained, the model could perform registration orders of magnitude faster than classical optimization-based methods. This made learning-based registration attractive for large-scale studies and time-sensitive pipelines [3, 8].

3.3.3 Architectural Characteristics

CNN-based registration models rely on convolutional kernels to extract hierarchical local features. Their main strength lies in strong locality priors: nearby voxels are processed in a way that naturally respects local anatomical structure. Encoder-decoder designs further allow the model to combine fine spatial detail with global contextual features through skip connections. These properties make CNNs especially effective for mono-modal deformable registration in volumetric medical images [8].

However, CNNs also have limitations. Their receptive field grows only gradually with depth, which makes it difficult to model long-range correspondence explicitly. This can be problematic when deformation is large or when the relationship between distant anatomical regions must be considered jointly. As a result, later work introduced hybrid and attention-based architectures to overcome these limitations [3, 9].

3.3.4 Strengths and Limitations

The main advantage of CNN-based registration is inference speed. After training, registration can be performed with a single forward pass. CNN-based methods also scale well to large datasets and integrate naturally into end-to-end medical image analysis pipelines.

Their limitations include dependence on training data, sensitivity to domain shift, and limited interpretability. A CNN trained on one dataset or scanner distribution may not generalize reliably to another. Moreover, although the output deformation may optimize the training loss, it is not automatically guaranteed to be anatomically plausible unless appropriate regularization or diffeomorphic constraints are included [3, 8].

3.4 Transformer-Based Registration

3.4.1 Motivation

Transformer-based registration methods were developed to address one of the major limitations of CNNs: the lack of explicit modeling of long-range spatial relationships. In image registration, correspondence is not always local. A deformation field may depend on anatomical context spanning a large region of the image, especially in inter-subject registration or in cases with large displacement. Self-attention mechanisms are well suited to this problem because they allow each token or patch to interact directly with many others [9].

3.4.2 Self-Attention Mechanism

At the core of Transformer models is the attention operation:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V, \tag{3.7}$$

where Q, K, and V are the query, key, and value matrices, and d_k is the key dimension. In registration, this mechanism enables the model to capture global anatomical dependencies rather than relying only on stacked local convolutions. As a result, Transformer-based methods can model long-range correspondences more explicitly than CNN-only architectures [9].

3.4.3 Representative Method: TransMorph

A representative Transformer-based registration method is **TransMorph**. TransMorph combines a Transformer-style encoder with a convolutional decoder in a U-shaped framework for unsupervised deformable medical image registration [9]. The Transformer encoder extracts global contextual features through self-attention, while the decoder reconstructs a dense deformation field at the original image resolution.

Like VoxelMorph, TransMorph is trained with an unsupervised objective combining image similarity and deformation regularization. Therefore, the key innovation is not a fundamentally different loss function, but a richer feature representation. By incorporating long-range attention, TransMorph improves the ability to capture complex correspondence patterns that are difficult for purely convolutional backbones [3, 9].

3.4.4 Why Transformers Matter in Registration

Transformers are particularly relevant in registration because anatomical structures are spatially distributed and often exhibit long-range dependency. For example, in brain registration, the correspondence of one cortical region may be informed by large-scale structural context. In inter-subject or multimodal registration, global anatomical layout can be just as important as local texture. Transformer models provide a natural framework for integrating this type of information [3, 9].

In practice, many Transformer-based registration methods are hybrid rather than purely attention-based. This reflects an important lesson from medical imaging: local detail is still crucial, so convolutional priors remain useful. The most successful models therefore tend to combine convolutional locality with Transformer-style global context rather than discarding CNN inductive bias entirely [3, 9].

3.4.5 Strengths and Limitations

Transformer-based methods often improve registration accuracy by capturing long-range spatial dependency more effectively than CNNs. They are especially promising for complex deformable registration and have quickly become a major research direction in recent years.

Their limitations include heavier computational cost, larger memory usage, and greater dependence on data and training design. Because self-attention does not inherently encode locality as strongly as convolution does, Transformer models may require more data or additional architectural constraints to achieve stable performance. For this reason, hybrid CNN–Transformer designs remain the dominant practical choice in current medical image registration research [3, 9].

3.5 Comparative Discussion of the Four Method Families

The four families considered in this report reflect different trade-offs rather than a simple chronological replacement of older methods by newer ones.

Classical optimization-based methods are explicit, interpretable, and often robust across domains, but they are computationally expensive because optimization is repeated for each image pair. Metaheuristic methods offer stronger global search capability and can better handle irregular objective landscapes, but they are usually even more computationally demanding and are difficult to scale to dense deformable registration. CNN-based methods greatly reduce inference time by learning an amortized mapping from image pairs to deformation fields, but they may suffer from domain shift and limited long-range modeling. Transformer-based methods address some of these representational limitations by introducing global self-attention, though at the price of greater complexity and computational burden [3, 8–10].

For this reason, no single family is universally optimal. Classical methods such as SyN remain strong baselines, especially when data are limited or cross-domain generalization is critical. CNN-based models are attractive for efficient large-scale deployment. Transformer-based models are promising when complex spatial relationships must be modeled more explicitly. Metaheuristics remain useful in specialized scenarios where global optimization is more important than runtime efficiency. Understanding these trade-offs is essential for selecting an appropriate method for a given biomedical registration problem.

3.6 Summary

This chapter reviewed four representative methodological directions in biomedical image registration. Classical optimization-based methods solve an explicitly defined energy function for each image pair and remain foundational for the field. Metaheuristic methods reformulate registration as a global search problem and are particularly useful when the objective landscape is highly non-convex. CNN-based methods such as VoxelMorph replace iterative test-time optimization with fast learned deformation prediction, while Transformer-based methods such as TransMorph further improve representation by modeling long-range spatial dependencies through self-attention. Together, these four families provide a coherent framework for understanding both the history and the current state of biomedical image registration.

Chapter 4

Datasets

4.1 Overview

A meaningful review of biomedical image registration methods should not rely on only one anatomical setting, because the difficulty of registration depends strongly on the organ, the imaging protocol, and the type of motion involved. For this reason, this report focuses on two representative public datasets: **OASIS** [16] for brain MRI registration and **DIR-Lab 4DCT** [17] for lung CT registration. The first provides a relatively stable neuroanatomical setting that is widely used in inter-subject deformable registration, while the second represents a harder scenario involving large respiratory motion and complex thoracic deformation [16–18].

The choice of OASIS and DIR-Lab is motivated by both methodological relevance and diversity of difficulty. OASIS is a widely used public neuroimaging dataset and a standard benchmark in atlas-based registration, morphometry, and learning-based registration pipelines. In contrast, DIR-Lab 4DCT is a well-known thoracic benchmark for deformable registration under respiratory motion, where large non-rigid displacement and sliding boundaries make the problem more challenging [5, 8, 9, 16–18].

Using these two datasets allows the report to cover two complementary settings. OASIS is suitable for studying dense deformable registration under relatively smooth anatomical variation, while DIR-Lab highlights the limitations of overly smooth deformation models and the need for robustness under large physiological motion. Together, they provide a more informative basis for discussing the strengths and weaknesses of the four method families considered in this report.

4.2 OASIS: Brain MRI Registration Benchmark

The *Open Access Series of Imaging Studies* (OASIS) is a landmark public neuroimaging initiative designed to make high-quality brain MRI data broadly available for research. The original OASIS-1 release consists of a cross-sectional collection of 416 subjects aged 18 to 96 years, with 434 MR sessions in total. For each subject, three or four T1-weighted MRI scans acquired within a single imaging session are provided. The cohort includes both cognitively normal and demented older adults, making the dataset valuable not only for registration research but also for morphometric and clinical neuroscience studies [16].

In the context of image registration, OASIS is particularly attractive for several reasons. First, it is a mono-modal brain MRI dataset, so the similarity relationship between images is more stable than in multimodal settings. Second, it supports inter-subject deformable registration, which is central to atlas construction, population analysis, and anatomical label propagation. Third, because the brain is enclosed by the skull and does not undergo large physiological deformation like thoracic organs, the deformation field is typically smoother and more anatomically constrained than in lung or abdominal registration. This makes OASIS an appropriate dataset for evaluating both classical methods such as SyN and modern deep learning approaches such as VoxelMorph and TransMorph [5, 8, 9].

Another practical advantage of OASIS is its prevalence in the literature. Many modern registration papers report results on OASIS directly or use closely related brain MRI benchmarks such as IXI and LPBA40. Consequently, OASIS provides a convenient common ground for comparing different methodological families under a well-studied and reproducible setting. For a report focused on fundamental concepts and representative methods, OASIS therefore serves as an ideal brain registration benchmark.

4.3 DIR-Lab 4DCT: Lung Registration Benchmark

The DIR-Lab 4DCT dataset, now distributed through the Emory Deformable Image Registration Laboratory, is one of the most influential benchmarks for thoracic deformable image registration. It contains 10 thoracic 4DCT cases acquired during radiotherapy planning, with each case consisting of 10 CT volumes sampled across different respiratory phases. The dataset includes expert landmark annotations between key phases, especially between end-inhalation and end-exhalation images, and also provides propagated landmark subsets for intermediate phases. The dataset is therefore highly suitable for evaluating spatial registration accuracy using target registration error (TRE) rather than relying only on overlap-based metrics [17, 18].

DIR-Lab is widely regarded as a difficult benchmark because thoracic registration is dominated by large respiratory motion, strong local deformation, and nonuniform displacement across the lung field. In addition, lung motion interacts with anatomical structures such as the diaphragm, chest wall, and vasculature, creating deformation patterns that are more complex than those commonly observed in brain MRI. This makes DIR-Lab especially useful for testing whether a registration method can remain accurate under large deformation while preserving geometric plausibility [17, 18].

An important feature of DIR-Lab is the availability of manually identified anatomical landmarks, which enable direct and interpretable evaluation of registration accuracy. For medical image registration, such benchmark landmarks are extremely valuable because dense ground-truth deformation fields are usually unavailable in real clinical data. In this sense, DIR-Lab complements OASIS: whereas OASIS is well suited for studying smooth inter-subject anatomical alignment and overlap-based evaluation, DIR-Lab emphasizes physically meaningful pointwise accuracy under demanding motion conditions [17, 18].

4.4 Datasets Comparison

The main characteristics of the two datasets are summarized in Table 4.1.

Table 4.1: Summary of OASIS and DIR-Lab

Aspect	**OASIS**	**DIR-Lab**
Anatomy	Brain	Lung / thorax
Modality	T1 MRI	4D CT
Data type	3D MRI	4D CT sequence
Scale	416 subjects, 434 sessions	10 cases
Per case	3–4 scans / subject	10 respiratory phases / case
Setting	Inter-subject registration	Intra-subject registration
Deformation	Smooth anatomical variation	Large respiratory motion
Annotation	Labels / overlap-based evaluation	Manual landmarks
Typical metric	Dice	TRE
Landmarks	No dense ground truth	300 landmark pairs / case
Difficulty	Moderate, more controlled	Hard, more realistic
Typical use	Brain MRI benchmarks	Lung motion benchmarks

The most important difference between the two datasets lies in the deformation pattern. OASIS mainly reflects relatively smooth anatomical variation across subjects, so it is suitable for controlled evaluation of deformable registration methods. In contrast, DIR-Lab involves large respiratory motion within the same subject, making the deformation more complex and harder to model. Another key difference is the evaluation protocol: OASIS is often assessed indirectly through overlap-based metrics such as Dice, while DIR-Lab provides manual landmarks and is therefore more suitable for direct accuracy evaluation using TRE. As a result, OASIS is better for standardized comparison, whereas DIR-Lab is more useful for testing robustness under realistic large deformation.

Chapter 5

Experiments

Chapter 6

Conclusion

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